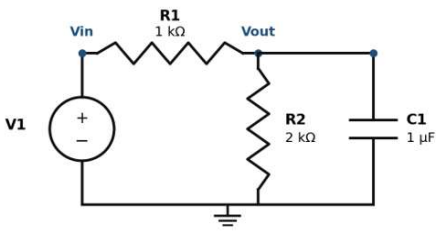
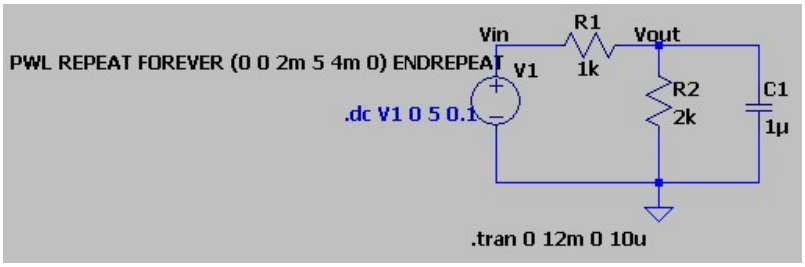
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4. Analyze the following first-order RC network using DC Sweep and Transient Analysis. Use both PULSE and PWL voltage-source excitations

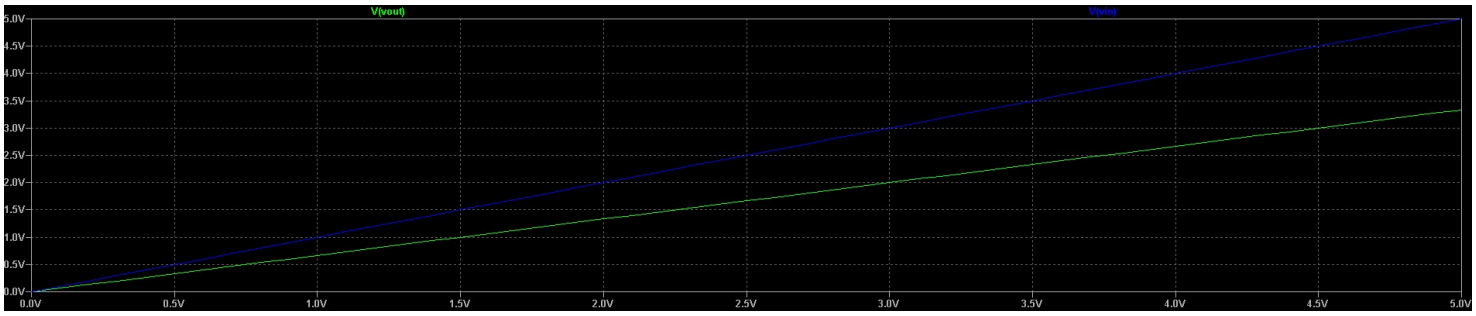




A. DC Sweep: Set V1 as a DC source and sweep it from 0 V to 5 V. Plot V(vout) as a function of the source voltage and compare the simulated result with the resistive-divider calculation. Use .dc V1 0 5 0.1

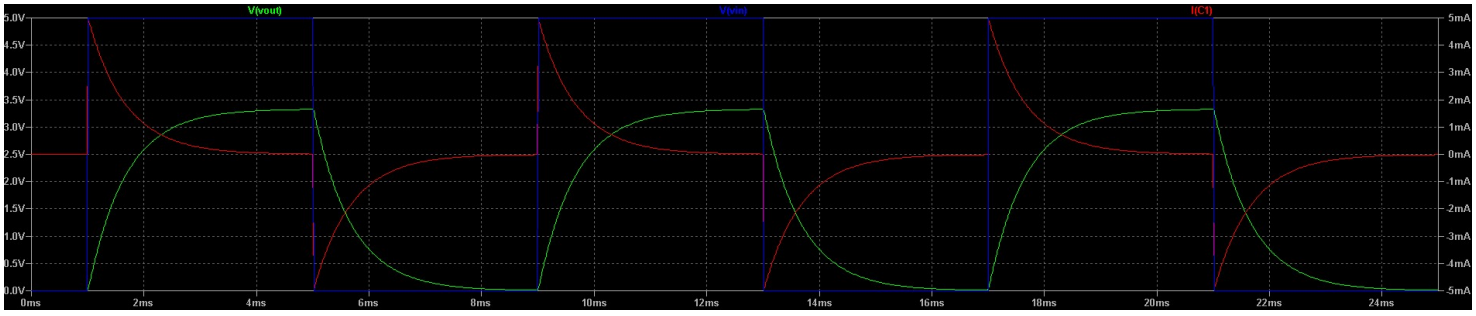
$V_{in} := 5\ \text{volt}$ $R_1 := 1\ \text{kohm}$ $R_2 := 2\ \text{kohm}$

$$V_{out} := V_{in} \cdot \left(\frac{R_2}{R_1 + R_2} \right) = 3.3333\ \text{V}$$



B. Pulse response. Replace the DC source with the PULSE definition below and run a transient simulation. Plot V(vin), V(vout), and I(C1).

Source definition: PULSE(0 5 1m 10u 10u 4m 8m)
Simulation directive: .tran 0 25m 0 10u



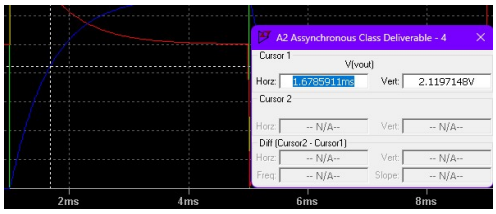
Calculate the expected time constant τ = (R1 || R2)·C1. Using the LTspice cursors, estimate the charging and discharging time constants from the waveform and compare them with the theoretical value. Also explain why the final value of Vout is not 5 V.

$C_1 := 1\ \mu\text{F}$

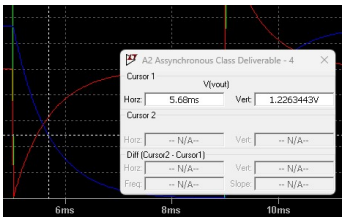
$$\tau := \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} \cdot C_1 = 0.0007\ \text{s}$$

According to $v_c(\tau) := \left(V_f + \left(V_0 - V_f \right) \cdot \mathbf{e}^{-1} \right)$

$v_{0_t} := 0.632 \cdot V_{out} = 2.1067\ \text{V}$



$v_{3.3_t} := 0.368 \cdot V_{out} = 1.2267\ \text{V}$



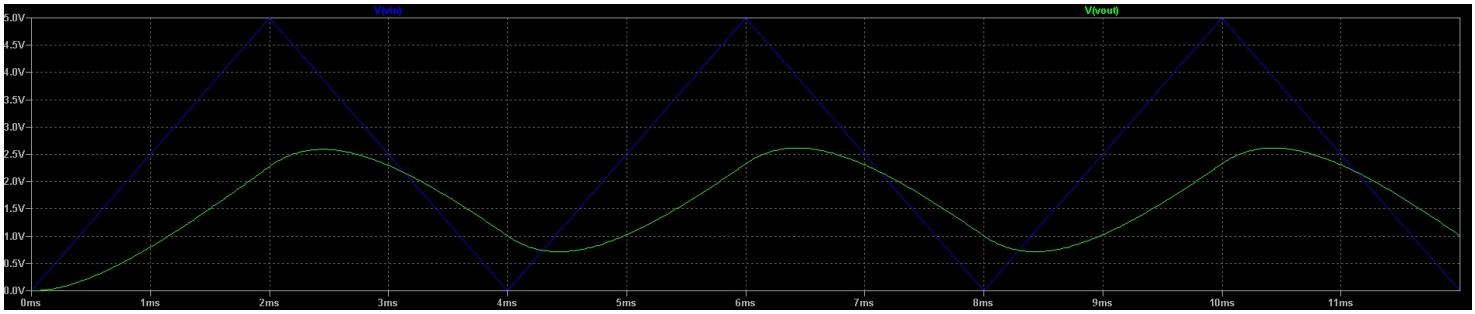
Vout value is not 5v because the Vin voltage is divided between R1 and R2.

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C. PWL excitation. Replace the PULSE source with the repeating triangular waveform below and simulate for at least three periods. Plot V(vin) and V(vout) on the same graph and describe the attenuation and time delay introduced by the RC network.

Source definition: PWL REPEAT FOREVER (0 0 2m 5 4m 0) ENDREPEAT
Suggested transient analysis: .tran 0 12m 0 10u



The RC network attenuates and delays the output because the capacitor voltage cannot change instantaneously. The resistive divider reduces the voltage, while the capacitor requires time to charge and discharge through the resistors.

The rounded corners show the low-pass filtering effect of the RC network, which smooths rapid changes in the input.

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